Nonradial mass-critical biharmonic NLS

Verified obstructions and partial finite-time blowup theorems

Research working note

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Abstract

We investigate whether every negative-energy $H^2(\mathbb{R}^N)$ solution of the focusing mass-critical biharmonic nonlinear Schrödinger equation blows up in finite time, without radial symmetry. The unrestricted question is not resolved here. We prove that the proposed unrestricted small-coefficient weighted interpolation estimate is false, even at fixed mass, energy -1, and zero momentum, and even with an optimally chosen localization center. We replace it by a weighted estimate involving mass in relative balls. Combined with the exponential virial identities, this gives a conditional blowup criterion and unconditional consequences for several invariant, genuinely nonradial symmetry classes. A putative global negative-energy solution must develop a precise form of angular concentration. A separate angular-regularity criterion and traveling-wave tests identify further constraints. The general virial mechanism is inherited from prior work; no external priority or peer-review claim is made for the additional derivations.

1 Problem, status, and provenance

Set $N \geq 2$ and $p = 2 + 8/N$. The equation and conserved quantities are

$$i\partial_t u = \Delta^2 u - |u|^{p-2}u, \qquad u(0) = u_0 \in H^2(\mathbb{R}^N), \tag{1}$$

$$m = M(u) = \|u\|_2^2, \qquad E(u) = \frac{1}{2}\|\Delta u\|_2^2 - \frac{1}{p}\|u\|_p^p, \qquad P(u) = \operatorname{Im}\int \bar{u}\nabla u\, dx. \tag{2}$$

Let $(-T_-, T_+)$ be the maximal H^2 lifespan. The target is

$$E(u_0) < 0 \quad \Longrightarrow \quad T_+ < \infty \text{ and } T_- < \infty \tag{3}$$

for arbitrary initial data, with no symmetry or finite-variance hypothesis. Finite lifespan means divergence of the H^2 norm through the usual blowup alternative. Infinite-time growth of that norm is not sufficient to prove (3).

Boulenger–Lenzmann [1] obtained the radial finite-time/infinite-time alternative. The manuscript posted on 2 October 2026 by Dinh [2] proves finite-time blowup for radial data in every $N \geq 2$. Its Remark 1.4 explicitly retains the nonradial question. The present literature check did not locate a full resolution under (1)–(3). Results with an inhomogeneous coefficient, extra dispersion, or additional symmetry do not directly answer this formulation.

The statements below distinguish three levels: established source identities; derivations proved in this note; and unresolved requirements. The proofs of the local-mass estimate, orbit-packing consequence, and static obstructions were cross-checked by independent derivations during this research session. This is not a substitute for external mathematical review or a proof of publication novelty.

2 The part of the virial argument that needs no symmetry

For a real smooth multiplier ϕ with the required bounded derivatives, define

$$V_\phi(u) = 2\,\mathrm{Im}\int \bar{u}\nabla\phi\cdot\nabla u\,dx.$$

Direct differentiation, integration by parts, and (1) give

$$\begin{aligned} V_\phi' = {} & 8\,\mathrm{Re}\sum_{i,j,k}\int \phi_{jk}\bar{u}_{ij}u_{ik}\,dx - 4\,\mathrm{Re}\sum_{j,k}\int(\Delta\phi)_{jk}\bar{u}_j u_k\,dx \\ & - 2\int \Delta^2\phi|\nabla u|^2\,dx + \int \Delta^3\phi|u|^2\,dx - \frac{8}{N+4}\int \Delta\phi|u|^p\,dx. \end{aligned} \tag{4}$$

For H^2 solutions this is understood via smooth approximation, or as the integrated localized multiplier identity. Each displayed right-hand term is continuous in H^2 for fixed multiplier. The local theory supplies the corresponding approximation on compact lifespan intervals.
Fix a center c and $R>0$. Write $r=|x-c|$, $s=r/R$, and choose

$$h = e^{-s^4}, \quad w = 1-h, \quad \nabla\phi_{R,c} = (x-c)h, \quad D = \int w|D^2u|_F^2\,dx, \quad L = \int w|u|^p\,dx. \tag{5}$$

The quartic exponent in this weight is important. It is not the Gaussian weight e^{-s^2}. The multiplier Hessian is

$$D^2\phi_{R,c} = hI - 4s^4h\,e_r\otimes e_r.$$

Thus the exact principal and nonlinear decomposition is

$$V_R' = 16E - 8D - 32\mathcal{H} + B + \mathcal{L}, \tag{6}$$

where

$$\begin{aligned} \mathcal{H} &= \int s^4h|D^2u\,e_r|^2\,dx \ge 0, \\ B &= \frac{8}{N+4}\int [Nw+4s^4h]|u|^p\,dx, \qquad 0\le B\le 8L, \\ \mathcal{L} &= -4\,\mathrm{Re}\sum_{j,k}\int(\Delta\phi_{R,c})_{jk}\bar{u}_j u_k\,dx - 2\int\Delta^2\phi_{R,c}|\nabla u|^2\,dx + \int\Delta^3\phi_{R,c}|u|^2\,dx. \end{aligned}$$

We used $s^4e^{-s^4}\le 1-e^{-s^4}$ and the unweighted identity $\|D^2u\|_{L^2,F}^2 = \|\Delta u\|_2^2$.
Let $g=\sqrt{w}$. Near the origin g is a smooth multiple of r^2/R^2; hence

$$\begin{aligned} \|\Delta g\|_\infty &\le C_N R^{-2}, \\ |D^2\Delta\phi_{R,c}| + |\Delta^2\phi_{R,c}| &\le C_N R^{-2}g, \qquad |\Delta^3\phi_{R,c}| \le C_N R^{-4}. \end{aligned}$$

Integration by parts gives

$$\begin{aligned} \int g|\nabla u|^2\,dx &= -\,\mathrm{Re}\int g\bar{u}\Delta u\,dx + \frac{1}{2}\int\Delta g|u|^2\,dx \\ &\le \sqrt{N}\,m^{1/2}D^{1/2} + C_N R^{-2}m. \end{aligned} \tag{7}$$

Consequently, for every $\eta>0$,

$$|\mathcal{L}| \le \eta D + C_{N,\eta}R^{-4}m. \tag{8}$$

Finally $|\nabla\phi_{R,c}|^4 \le C_N R^4 w$, Cauchy–Schwarz, and (7) imply

$$|V_R(u)| \le C_N m + C_N R m^{3/4}D^{1/4}. \tag{9}$$

Equations (6)–(9) require no radial symmetry. They are the inherited exponential virial mechanism. The missing input is control of L.

3 Why direct unrestricted absorption cannot work

3.1 A translated concentration

For $a \neq c$, $f \in C_c^\infty(\mathbb{R}^N) \setminus \{0\}$, put

$$f_{\lambda,a}(x) = \lambda^{N/2} f(\lambda(x-a)).$$

Mass is independent of λ, while, with R, c, a fixed,

$$\lambda^{-4} \int w |f_{\lambda,a}|^p \, dx \longrightarrow w(a) \, \|f\|_p^p ,$$
$$\lambda^{-4} \int w |D^2 f_{\lambda,a}|_F^2 \, dx \longrightarrow w(a) \left\| D^2 f \right\|_2^2 .$$

Therefore an estimate $L \leq \varepsilon D + C_{\varepsilon,m} R^{-4}$ for arbitrary $\varepsilon > 0$ and all H^2 data would force $\|f\|_p^p \leq \varepsilon \|D^2 f\|_2^2$ for arbitrary ε, a contradiction.

3.2 Fixed invariants and an optimally chosen center

Proposition 3.1. *For every fixed $R > 0$ there are $m_0 > 0$ and real $U_\lambda \in C_c^\infty(\mathbb{R}^N)$ such that*

$$M(U_\lambda) = m_0, \qquad E(U_\lambda) = -1, \qquad P(U_\lambda) = 0,$$

and, uniformly over all centers $c \in \mathbb{R}^N$,

$$\frac{L_{R,c}(U_\lambda)}{D_{R,c}(U_\lambda)} \longrightarrow \frac{p}{2}, \qquad \inf_c D_{R,c}(U_\lambda) \longrightarrow \infty.$$

In particular, choosing a center separately for each datum cannot give $L_{R,c} \leq \varepsilon D_{R,c} + C$ for any $\varepsilon < p/2$, even if C depends on the fixed mass, energy, momentum, and R.

Proof. Choose real radial $q \in C_c^\infty \setminus \{0\}$ and $A > 0$ so that $f = Aq$ has zero energy. Explicitly,

$$A^{p-2} = \frac{p \, \|\Delta q\|_2^2}{2 \, \|q\|_p^p}.$$

Then $K := \|D^2 f\|_2^2 > 0$ and $Q := \|f\|_p^p = (p/2)K$. Choose a real compactly supported b with $E(b) = -1$: first increase amplitude to make energy negative, then use critical spatial scaling to normalize it. Choose distinct points a_1, a_2 outside the support of b and put

$$U_\lambda = f_{\lambda,a_1} + f_{\lambda,a_2} + b.$$

For all large λ the three supports are disjoint. Thus the invariant identities hold exactly, with $m_0 = 2M(f) + M(b)$.
Set $W_c = w_{R,c}(a_1) + w_{R,c}(a_2)$ and $d = |a_1 - a_2|$. At least one point is at distance $d/2$ or more from c, so

$$W_c \geq 1 - e^{-(d/(2R))^4} > 0.$$

The translated weights satisfy $\|\nabla w_{R,c}\|_\infty \leq C/R$ uniformly in c. Changing variables and using the mean value theorem yields

$$L_{R,c}(U_\lambda) = \lambda^4 W_c Q + O_R(\lambda^3) + O(1), \quad D_{R,c}(U_\lambda) = \lambda^4 W_c K + O_R(\lambda^3) + O(1),$$

with both remainders uniform in c. Division proves the claim. □

This proposition is an obstruction to a static estimate, not a global solution and not a counterexample to (3). It does not rule out a scale R that changes with the datum or time, provided the resulting new terms can be controlled.

3.3 Keeping the signed terms does not recover the same defect

Let $\mathcal{V}_\phi(u)$ denote the right-hand side of (4). For the real radial zero-energy f above,

$$\mathrm{Re}\sum_i \int \bar{f}_{ij} f_{ik}\, dx = \frac{K}{N}\delta_{jk}.$$

The leading λ^4 contribution of one packet to $\mathcal{V}_\phi$ is

$$\lambda^4 \Delta\phi(a)\left(\frac{8K}{N} - \frac{8Q}{N+4}\right) = 0.$$

The error is $O_R(\lambda^3)$ for translated exponential multipliers; lower-order terms are $O_R(\lambda^2)$. Applying this to both packets in Proposition 3.1 shows that a universal estimate

$$\mathcal{V}_{\phi_{R,c}}(u) \le 16E(u) - \gamma D_{R,c}(u) + C, \qquad \gamma > 0,$$

also fails on this fixed-invariant class, even with optimized c. The defect grows as λ^4, while $\mathcal{V}$ has no contribution of that order. Rearranging the same signed terms is therefore insufficient.

4 A valid local-mass replacement

For $0 < \delta < 1/4$ and fixed c, define

$$\mu_{\delta,c}(u) = \sup_{a \ne c} \int_{B(a,\delta|a-c|)} |u(x)|^2\, dx. \tag{10}$$

These balls are narrow relative to their distance from the center. This quantity measures concentration away from a single distinguished point.

Lemma 4.1. *There is C_N independent of u, R, c, δ such that*

$$L_{R,c}(u) \le C_N \mu_{\delta,c}(u)^{4/N}\left[D_{R,c}(u) + \delta^{-4}R^{-4}M(u)\right], \qquad u \in H^2(\mathbb{R}^N). \tag{11}$$

Proof. First work on $B(a,r)$. Put $m_a = \int_{B(a,4r)} |u|^2$ and $H_a = \int_{B(a,4r)} |D^2u|_F^2$. Choose smooth cutoffs $\eta = 1$ on $B(a,r)$ supported in $B(a,2r)$ and $\chi = 1$ on $B(a,2r)$ supported in $B(a,4r)$. Their derivatives have the natural powers of r^{-1}. Writing $J = \int \chi^2|\nabla u|^2$, integration by parts gives

$$J \le \sqrt{N}\, m_a^{1/2} H_a^{1/2} + C_N r^{-1} m_a^{1/2} J^{1/2}.$$

Absorption and Young's inequality yield

$$r^{-2}\int_{B(a,2r)} |\nabla u|^2 \le C_N H_a + C_N r^{-4} m_a.$$

The product rule and the mass-critical Gagliardo–Nirenberg inequality imply

$$\int_{B(a,r)} |u|^p \le \|\eta u\|_p^p \le C_N m_a^{4/N}(H_a + r^{-4}m_a). \tag{12}$$

Translate so $c = 0$. On each annulus $A_j = \{2^j \le |x| < 2^{j+1}\}$, choose a maximal r_j-separated family of centers with $r_j = \delta 2^j/64$. Balls of radius r_j cover A_j. Balls with radius $4r_j$ have bounded overlap, with a bound depending only on N, both within each annulus and across all annuli. Each enlarged ball is contained in $B(a, \delta|a|)$, so $m_a \le \mu_{\delta,0}(u)$.

On every enlarged ball, $w_R(x)$ is comparable to $w_R(a)$ with constants independent of δ, R, j. This follows from $w_R(x) \asymp \min\{|x|^4/R^4, 1\}$. Also

$$w_R(a) r_j^{-4} \le C_N \delta^{-4} R^{-4}.$$

Multiply (12) by $w_R(a)$ and sum over the cover. Bounded overlap controls the Hessian sum by $D_{R,0}$ and the cutoff sum by $C_N\delta^{-4}R^{-4}m$. This proves (11). The cutoff argument is valid directly in H^2, or by smooth approximation. □

5 Conditional finite-time blowup and a necessary concentration pattern

Theorem 5.1. *There exists $m_*(N) > 0$ such that a maximal solution with $E(u_0) < 0$ has $T_+ < \infty$ if some fixed c and some fixed $\delta \in (0, 1/4)$ satisfy*

$$\sup_{0 \le t < T_+} \mu_{\delta,c}(u(t)) \le m_*(N). \tag{13}$$

An analogous backward-time statement holds.

Proof. Take $\eta = 2$ in (8), and choose m_* so that the coefficient $C_N m_*^{4/N}$ in (11) is at most $1/4$. Then

$$V_R' \le 16E - 4D + C_N R^{-4} m (1 + \delta^{-4} m_*^{4/N}).$$

Choose one sufficiently large, fixed R to obtain

$$V_R' \le -8|E| - 4D. \tag{14}$$

Suppose the solution were forward global. With $y = -V_R$, (14) makes y reach $2C_N m$ after finite time. Thereafter (9) implies

$$D \ge c_N R^{-4} m^{-3} y^4, \qquad y' \ge c_N R^{-4} m^{-3} y^4.$$

Integrating the differential inequality for y^{-3} gives a contradiction in finite time. Backward time uses $\widetilde{u}(t,x) = \overline{u(-t,x)}$, which satisfies the same equation with the same mass, energy, and relative-ball masses. □

The hypothesis is dynamical. It cannot be inferred from negative energy alone by the estimates established here.

Corollary 5.2. *If a negative-energy solution is forward global, then for every fixed c there exist $t_n \to \infty$, $\delta_n \downarrow 0$, and $a_n \ne c$ such that*

$$\int_{B(a_n, \delta_n |a_n - c|)} |u(t_n, x)|^2 \, dx \ge \frac{1}{2} m_*(N).$$

Proof. For every fixed $f \in L^2$, $\mu_{\delta,c}(f) \to 0$ as $\delta \downarrow 0$. Indeed, centers very far from c give balls in a small spatial tail, while centers in a bounded region give sets of uniformly vanishing volume. Absolute continuity of the mass integral handles the latter. Moreover

$$|\mu_{\delta,c}(f) - \mu_{\delta,c}(g)| \le (\|f\|_2 + \|g\|_2) \, \|f - g\|_2 .$$

The convergence is therefore uniform on compact subsets of L^2, hence on each compact time interval of the solution. The contrapositive of Theorem 5.1 supplies, for every δ, a time and a relative ball carrying more than $m_*/2$. Choose δ_n recursively smaller than 2^{-n} and $\delta_{n-1}/2$, and so small that this cannot happen before time n; then choose such a ball. No attainment of the supremum is needed. □

This identifies a scenario that a general proof must exclude or control. It does not itself exclude that scenario. Negative energy supplies concentration somewhere; it does not supply the uniform upper bound in (13).

6 Unconditional results in invariant nonradial classes

Theorem 6.1 (Infinite-orbit symmetry)**.** *Let $G \subset O(N)$ be compact, with Gx infinite for every $x \neq 0$. Let $\chi : G \to S^1$ be a continuous unitary character. Suppose*

$$u_0(gx) = \chi(g)u_0(x), \qquad g \in G, \qquad u_0 \in H^2(\mathbb{R}^N), \quad E(u_0) < 0.$$

Then the maximal solution has $T_+, T_- < \infty$.

Proof. Rotation invariance, phase invariance, and uniqueness preserve the equivariance at every time. In particular the solution's mass density is G-invariant.

Fix any integer K. Every unit-vector orbit contains K distinct points. For each unit vector choose K group elements realizing positive minimum pairwise separation. The same separation, with a factor $1/2$, holds in a neighborhood of that vector. A finite cover of the unit sphere gives a uniform $d_K > 0$ such that every orbit has K points separated by at least $d_K|x|$.

For $\delta < \min\{1/4, d_K/2\}$ the corresponding K balls of radius $\delta|x|$ are disjoint and have equal mass. Hence $\mu_{\delta,0}(u(t)) \leq m/K$ at every time. Choose K so large that $m/K \leq m_*(N)$ and apply Theorem 5.1 in both time directions. □

Only initial *density* symmetry is not enough: it need not be preserved. The theorem imposes symmetry on the complex field, with a compatible phase character.

Planar vortices. In $N = 2$, take $G = SO(2)$ and $\chi(R_\theta) = e^{i\ell\theta}$, $\ell \in \mathbb{Z}$. This gives the rigorous implication

$$u_0(r, \theta) = e^{i\ell\theta}v_0(r) \in H^2(\mathbb{R}^2), \quad E(u_0) < 0 \quad \Longrightarrow \quad T_+, T_- < \infty.$$

For $\ell \neq 0$ the complex field is nonradial. The class is nonempty: multiply $(x_1 + ix_2)^\ell e^{-|x|^2}$ for $\ell \geq 0$, or $(x_1 - ix_2)^{|\ell|}e^{-|x|^2}$ for $\ell < 0$, by a sufficiently large positive amplitude. This remains a restricted class and does not address arbitrary planar data.

Biradial data in all $N \geq 4$. Take $G = O(2) \times O(N - 2)$ and write $x = (y, z)$. Then $u_0(x) = v_0(|y|, |z|)$ is allowed. Every nonzero orbit is infinite because at least one block has nonzero length and both block dimensions are at least two. A genuinely nonradial example is $u_0 = Ae^{-|y|^2-2|z|^2}$ for sufficiently large A. No finite-variance hypothesis is needed in the theorem, even though this example has finite variance.

Finite cyclic symmetry. For $N = 2d$ and an integer $K \geq 2$, identify $\mathbb{R}^N = \mathbb{C}^d$ and put $g(z) = e^{2\pi i/K}z$. If $u_0(gx) = \zeta u_0(x)$ with $\zeta^K = 1$, then for $\delta < \min\{1/4, \sin(\pi/K)\}$ the K orbit balls are disjoint. Thus

$$E(u_0) < 0, \qquad M(u_0)/K \leq m_*(N) \quad \Longrightarrow \quad T_+, T_- < \infty.$$

Here only a finite symmetry is imposed, but a mass-to-orbit-size condition is required. These statements are consequences proved in this note; their external novelty has not been established.

7 A complementary angular-regularity criterion

Define, when finite,

$$K(u) = \int_0^\infty \|(1 - \Delta_{S^{N-1}})u(r, \cdot)\|^2_{L^2(S^{N-1})}\, r^{N-1}\, dr.$$

This angular H^2 mass satisfies $K \geq m$. It is not automatically finite for arbitrary $H^2(\mathbb{R}^N)$ data and is not generally conserved.

Proposition 7.1. *For every $\varepsilon > 0$,*

$$L_{R,0}(u) \le \varepsilon D_{R,0}(u) + C_{N,\varepsilon} R^{-4} \left[K(u) m^{4/(N-1)} + K(u)^{(N-1)/N} m^{5/N} \right]. \tag{15}$$

Consequently a global negative-energy solution cannot satisfy

$$\limsup_{T\to\infty} \frac{1}{T} \int_0^T K(u(t))\, dt < \infty.$$

Proof. On a fixed annular cylinder, let v be a localized rescaled function and set $\mu = \|v\|_2^2$, $Q = \|\partial_r^2 v\|_2^2 + \mu$, and $A = \|v\|_{L_r^2 H_\theta^2}^2$, using equivalent norms with $A \ge \mu$. The anisotropic inequality is

$$\|v\|_p^p \le C_N Q^{1/N} A^{(N-1)/N} \mu^{4/N}. \tag{16}$$

To verify it, use finitely many smooth sphere charts and Fourier variables $(\xi, \eta) \in \mathbb{R} \times \mathbb{R}^{N-1}$. Hausdorff–Young and weighted Hölder with $W = 1 + (\xi/a)^4 + |\eta/b|^4$ give

$$\|v\|_p \le C \left(\int W |\widehat{v}|^2 \right)^{1/2} \left(\int W^{-p/(p-2)} \right)^{1/2-1/p}.$$

The integral is finite since $p/(p-2) = (N+4)/4 > N/4$. Put $a = (Q/\mu)^{1/4}$ and $b = (A/\mu)^{1/4}$; the case $\mu = 0$ is immediate. Raising to the power p gives (16). Chart and interval cutoff derivatives are controlled by their fixed-domain H^2 norms.

On dyadic annuli of radius $\rho_j = 2^j R$, let $\alpha_j = \rho_j^{-4} \min\{1, (\rho_j/R)^4\} \le R^{-4}$. Write D_j, A_j, μ_j for the weighted Hessian, angular mass, and mass on fixed enlarged annuli. Rescaling and the cutoff gradient estimate used in Lemma 4.1 give

$$\begin{aligned} \int_{A_j} w|u|^p \le{}& C\alpha_j^{1-1/N} D_j^{1/N} A_j^{(N-1)/N} \mu_j^{4/N} \\ &+ C\alpha_j A_j^{(N-1)/N} \mu_j^{5/N}. \end{aligned}$$

Young's inequality bounds the first term by $\varepsilon' D_j + C_{\varepsilon'} \alpha_j A_j \mu_j^{4/(N-1)}$. Sum with bounded overlap and use

$$\sum_j A_j^{(N-1)/N} \mu_j^{5/N} \le \left(\sum_j A_j \right)^{(N-1)/N} \left(\sum_j \mu_j^5 \right)^{1/N}.$$

This proves (15).

For the time-average assertion, $K \ge m$ implies $K^{(N-1)/N} m^{5/N} \le K m^{4/N}$. Combine (15) with the virial bounds, integrate in time, and choose one fixed large R using the supposed finite upper average of K. For $y = -V_R$ this gives

$$y(t) \ge 8|E|t - C + c \int_0^t D(s)\, ds.$$

The bridge (9) implies $D \ge c_R (y - C_N m)_+^4$. The right-hand integral lower bound therefore satisfies a superlinear integral differential inequality once positive. More explicitly, define $F(t) = 8|E|t - C + c\int_0^t D$; then $y \ge F$, and for all sufficiently large t, $F' \ge c_R (F - C_N m)^4$. This contradicts global existence. □

For a planar ℓ-vortex, $K = (1+\ell^2)^2 m$ is constant; this independently recovers that case of Theorem 6.1. Conserved total angular momentum, a much weaker signed quantity, cannot replace K.

Corollary 7.2 (Quantitative angular growth)**.** *If a forward-global negative-energy solution has locally integrable $K(t) = K(u(t))$, then there exist $c(u_0, N) > 0$ and T_0 such that*

$$\int_0^T K(t)\,dt \geq c(u_0, N)T^5, \qquad T \geq T_0.$$

Proof. Put $e = |E|$ and $S(T) = \int_0^T K$. The preceding estimates, using $K \geq m > 0$, give

$$y(t) \geq -V_R(u_0) + 16et + c_N \int_0^t D(s)\,ds - C_{N,m}R^{-4}S(t).$$

Also $|V_R(u_0)| \leq C(u_0)R$ by Cauchy–Schwarz and boundedness of $\nabla\phi_R/R$. For each large test horizon T, choose $R = \alpha T$, which remains fixed throughout $[0, T]$. First choose $\alpha > 0$ small enough that $C(u_0)\alpha \leq e$. If $S(T) \leq \kappa T^5$, choose $\kappa > 0$ small enough that $C_{N,m}\alpha^{-4}\kappa \leq e$. Then, for $T/2 \leq t \leq T$,

$$y(t) \geq 6eT + c_N \int_{T/2}^t D(s)\,ds =: F(t).$$

For sufficiently large T, $F \geq 2C_N m$. The bridge gives $F' \geq c_{N,m}\alpha^{-4}T^{-4}F^4$. Starting at $F(T/2) = 6eT$, this forces divergence within $C_{N,m,e}\alpha^4 T$. Choosing α smaller, before choosing κ, makes this time less than $T/2$, a contradiction. Thus $S(T) > \kappa T^5$ for every sufficiently large T. No derivative of a time-dependent R was used. □

8 Additional proof obstructions and counterexample tests

8.1 A direct fixed-multiplier Riccati inequality can have the wrong sign

It is also possible to have very negative V_R but very positive instantaneous V_R' at fixed mass, energy -1, and momentum zero. Here is a reproducible algebraic construction.
Choose real radial $q \in C_c^\infty \setminus \{0\}$, $k \neq 0$, and $A > 0$ so $f = Ae^{ik\cdot x}q$ has zero energy. Put $m_q = \|q\|_2^2$, $a_q = \|\nabla q\|_2^2/N$, and $b_k = (N+4)a_q + |k|^2 m_q$. The traceless integrated kinetic tensor is

$$\mathrm{Re}\sum_i \int \bar{f}_{ij} f_{ik} - \frac{\|\Delta f\|_2^2}{N}\delta_{jk} = A^2 b_k \left(k_j k_k - \frac{|k|^2}{N}\delta_{jk}\right).$$

This identity follows by shifting the radial Fourier density by k and expanding its fourth moments. Place critically scaled copies at $\pm ae$, with modulations k and $-k$, where

$$a > 0, \quad |e| = 1, \qquad -|k|/\sqrt{N} < k \cdot e < 0.$$

More explicitly, add a fixed disjoint real packet b of energy -1 and set

$$U_\lambda(x) = A\lambda^{N/2}[e^{i\lambda k\cdot(x-ae)}q(\lambda(x-ae)) + e^{-i\lambda k\cdot(x+ae)}q(\lambda(x+ae))] + b(x).$$

The physical modulations are $\pm\lambda k$. For the multiplier centered at zero,

$$V_R(U_\lambda) = 4\lambda a e^{-a^4/R^4} A^2 m_q (k\cdot e) + O(1) \to -\infty.$$

Write $h_r = e^{-s}(1-4s)$ and $h_t = e^{-s}$ with $s = a^4/R^4$. Each concentrated packet contributes

$$8\lambda^4(h_r - h_t)A^2 b_k \left((k\cdot e)^2 - |k|^2/N\right) + O_R(\lambda^3) > 0$$

to the derivative for large λ. Thus $\mathcal{V}_{\phi_R}(U_\lambda) \to +\infty$. A universal instantaneous negative quartic inequality for this fixed V_R cannot hold without further information.

8.2 Moving centers and traveling profiles

If $c = c(t)$, differentiation adds the term

$$-2\dot{c}(t) \cdot \operatorname{Im} \int \bar{u}\, D^2 \phi_{R,c(t)} \nabla u \, dx.$$

It cannot simply be discarded. Total momentum zero does not force each separated component to have zero momentum. Fourth-order dispersion also lacks the usual second-order Galilean symmetry.

A possible genuine counterexample would be a traveling wave $u(t,x) = e^{i\omega t} Q(x - vt)$ with $Q \in H^2$ and $E(Q) < 0$. Its profile solves

$$\Delta^2 Q + \omega Q + iv \cdot \nabla Q - |Q|^{p-2} Q = 0.$$

For a sufficiently localized profile, differentiating $E + \omega M/2 - v \cdot P/2$ along mass-preserving scaling gives

$$4E(Q) - \tfrac{1}{2} v \cdot P(Q) = 0, \qquad E(Q) = \tfrac{1}{8} v \cdot P(Q).$$

Standing profiles have zero energy. This identity does not by itself determine the sign for arbitrary excited traveling profiles.

Exploratory one-dimensional and planar profile computations were undertaken in this session. A 14-pulse one-dimensional candidate at velocity $v = 0.01$ gave energy approximately -4.35117×10^{-7}, with $E - vP/8$ approximately -1.84×10^{-13} in a finite-domain boundary-value calculation. This is a candidate, not a certified real-line solution. Planar collinear and square multipulse tests did not produce a negative response form. These calculations do not answer the specified $N \geq 2$ problem. Domains, boundary conditions, solver limitations, data, and code are retained in the accompanying research bundle; the proved statements above do not depend on these computations.

9 What remains to solve the original problem

For arbitrary initial data, the missing step is a time-independent δ and fixed center satisfying (13), or another argument controlling the concentration in Corollary 5.2. The static obstructions show why conserved mass, energy, and momentum do not close the proposed multiplier argument. A complete proof must exploit additional dynamics, a different coercive functional, or controlled multi-center transport. A rigorous negative-energy global solution would instead disprove the conjecture. Neither conclusion has been obtained. **The unrestricted problem remains unresolved by this work;** the proved outputs are the obstruction statements, replacement estimates, and conditional and symmetry-restricted theorems above.

References

[1] T. Boulenger and E. Lenzmann, *Blowup for Biharmonic NLS*, arXiv:1503.01741, version 2, 21 April 2015. https://arxiv.org/abs/1503.01741.

[2] L. Dinh, *Finite-Time Blow-Up of Radial Negative-Energy Solutions for the Mass-Critical Biharmonic Nonlinear Schrödinger Equation*, 20-page manuscript, Meta AI Research, posted 2 October 2026. Official publication page and full PDF.